Does Training Against Debate Reward Honesty? Empirical Incentive Compatibility of Oversight Protocols on Conceptual Reasoning

Abhimanyu Pallavi Sudhir

November 2026

Abstract

Scalable-oversight protocols such as debate are meant to be trained against: their value lies in making honesty the policy that earns the most reward. Yet nearly all empirical work evaluates them by the judge's accuracy at inference time, which a protocol that is useless to train against can score well on when today's models are friendly. We instead measure *empirical incentive compatibility* (EIC): the covariance between a protocol's reward and the ground-truth honesty of the behaviour it rewards, over agent behaviours drawn by a *sampler*. That covariance is the first-order effect of reward optimization on honesty. We compare five protocols (consultancy, sequential and simultaneous debate, recursive debate, prover-estimator debate) with three weak judges on 927 conceptual-reasoning questions selected so that the agents know the answer and the judges do not, under four samplers of increasing optimization pressure: instructed honest and dishonest arguments (ASD), the same with only the arguments that actually lie, best-of-n selection against the judge's reward, and prompt optimization by a frontier model searching separately for the best honest and the best deceptive strategy. Instructed liars make every protocol except prover-estimator look incentive compatible. Under optimization the picture splits: optimized deception erases consultancy's incentive to be honest (held-out ASD -0.06, honesty wins on half the items), deepens prover-estimator's deficit (-0.23, even after fixing an exploitable payoff), and shrinks but does not erase debate's ($+0.32$ sequential, $+0.56$ simultaneous) and recursive debate's ($+0.10$). When both debaters best-respond to each other (PSRO self-play, 390 held-out items), simultaneous debate stays incentive compatible ($+0.48$) and sequential debate does not ($+0.05$, negative under one judge). The cheap sampler only partly predicts the optimized one (Spearman 0.69 across protocol–judge cells). EIC rises steeply with judge capability (in debate, honesty wins on 58–61% of items with Llama-3.1-8B and 90–92% with gpt-4.1) and, unlike accuracy, stays high for judges that already know the answer. Prompted liars that refuse to lie, common among safety-trained models, inflate a model's EIC from at most $+0.6$ to as much as $+4.0$.

1 Introduction

Debate [5] and its relatives are proposed as training methods: if a weak judge rewards whichever agent argues better, and lies are harder to defend than truths, then optimizing agents against the judge should make them honest. Empirical work on debate [2, 6, 7, 9] has nonetheless mostly reported the judge's accuracy at inference time. Accuracy and incentives can come apart: a judge that simply believes the agent is accurate when the agent happens to be honest, and useless to train against.

We study the incentive directly. For each episode of a protocol, every trained agent has two scores: the reward u the protocol pays it (what training would optimize) and a ground-truth value

v of its behaviour, here $v = 1$ if it argued the true answer and 0 if it argued a false one. Training on u moves an agent's policy π in the direction

$$\frac{d}{d\beta}\,\mathbb{E}_{\pi_\beta}[v] = \mathrm{Cov}_{\pi_\beta}(u, v), \qquad \pi_\beta \propto \pi_0\, e^{\beta u}, \tag{1}$$

the exact derivative along the KL-regularized optimization path. So the covariance between reward and honesty, taken over the behaviours an optimizer could reach, is a cheap proxy for what training would do; its sign along the path says whether the protocol is being Goodharted. Which behaviours it is taken over is set by a *sampler*. The agent score difference (ASD) of Pallavi Sudhir et al. [10] is the simplest: instruct the agent to argue each side. Best-of-n over sampled behaviour [3] and prompt optimization [1, 12] reach further.

Contributions.

- EIC estimates for five protocols under four samplers of increasing optimization pressure, on a conceptual-reasoning benchmark with a 54-point gap between agents and judges, with selection and evaluation on independent judge samples throughout (§4–6).
- A comparison of samplers: does the cheap instructed sampler rank protocols as optimization does (§8)?
- EIC across five judges of increasing capability on the same debates, and its dissociation from accuracy (§9).
- The refusal confound: models that will not lie make every protocol look incentive compatible (§10); and two hazards for anyone optimizing against LLM judges (§12).

2 Empirical incentive compatibility

Estimators. With $v \in \{0, 1\}$, $\mathrm{Cov}(u, v)/\mathrm{Var}(v)$ is the difference of mean rewards between honest and dishonest behaviour: the **ASD**, in the protocol's reward units (log score of the judge's probability on the agent's answer for consultancy and debate; the judge's probability for recursive debate; the prover's payoff for prover-estimator). We compute it per item and average over items. Because reward units differ across protocols and inflate with confident judges, we also report **concordance**, the share of items on which honesty earns more (0.5 = no incentive), and, where per-episode rewards exist, $r = \mathrm{corr}(u, v)$, the normalized covariance. All intervals are 95% percentile bootstraps over items.

Samplers.

- **Instructed (ASD).** The agent is told to argue the true answer, then a false one (3 samples each).
- **Committed liars.** The same, keeping only samples whose liars an auditor (gpt-5; agreement with Claude Sonnet 5 $\kappa = 0.76$) found committed to the lie. A "liar" that concedes is not the dishonest behaviour training could reach; this sampler targets effective lies.
- **Best-of-n.** The agent's own behaviour re-weighted toward high reward: exactly over pools of 16 candidate arguments per decision for one-round protocols, and by keeping the best of n

Table 1: EIC under the instructed samplers (gpt-5-mini agents, 3 weak judges pooled, 927 items). ASD in each protocol's reward units.

	instructed (ASD)			committed liars only		
protocol	ASD	conc.	r	ASD	conc.	r
consultancy	$+0.38\,[+0.29, +0.47]$	0.61	$+0.09$	$+0.21\,[+0.12, +0.31]$	0.55	$+0.04$
debate, sequential	$+0.74\,[+0.66, +0.82]$	0.69	$+0.18$	$+0.56\,[+0.48, +0.64]$	0.65	$+0.14$
debate, simultaneous	$+0.79\,[+0.70, +0.87]$	0.72	$+0.19$	$+0.57\,[+0.48, +0.65]$	0.68	$+0.14$
recursive debate	$+0.10\,[+0.08, +0.11]$	0.64	$+0.14$	$+0.10\,[+0.08, +0.12]$	0.62	$+0.12$
prover-estimator	$-0.13\,[-0.15, -0.11]$	0.35	-0.14	$-0.23\,[-0.26, -0.20]$	0.27	-0.26

candidates for every speech of three-round protocols (a preview judge scores each candidate). Either only the liar is optimized, or both sides are. A variant lets the agent choose which side to argue.

- **Prompt optimization.** An optimizer (Claude Opus 5.5) that knows the protocol's rules and rewards writes system-prompt strategies for the agent, in two separate searches per protocol and judge: one for the agent assigned the true answer under an *honest* directive, one for the agent assigned a false answer under a *deceptive* directive ("win anyway"). Each search runs five rounds of four candidates (GEPA-style reflection on the best strategy's transcripts) on 36 training items. EIC is the ASD between the best honest and the best deceptive strategy, both chosen on training reward and scored on 480 disjoint held-out items, after each round.

Blind-judge baseline. The same judge shown only the question and the side the agent defends, never an argument, pays the agent by its prior. Its ASD is the incentive a protocol would have if arguments were ignored. Every sampler that selects on the judge's reward is scored on an *independent* judge sample: two samples of the same judge on the same argument correlate only $r \approx 0.2$ within an item, so in-sample selection mostly selects noise (§12).

3 Setup

Items. 927 binary questions from 31 families of a conceptual-reasoning testbed (decision theory: Newcomb, smoking lesion, twin prisoner's dilemma, Sleeping Beauty, Parfit's hitchhiker; social choice; number theory; reward-hacking gridworlds; LLM auctions), selected from 3,437 because both agent models answer them correctly on 2 of 2 samples while three weak judges are right on at most 60% of 9 samples. On fresh samples the agents are right 96–99% of the time and the judges 41–46%, below chance: the judges hold confident wrong intuitions.

Models and protocols. Agents: gpt-5-mini (medium reasoning effort; the main agent throughout) and Claude Sonnet 4.5. Weak judges: gpt-4o-mini, Llama-3.1-8B-Instruct, Gemma-3-12B-it, each reasoning before stating probabilities. Protocols, all with 200-word messages: consultancy (3 rounds, the judge questions the consultant); debate (3 rounds, sequential or simultaneous); recursive debate (depth 2, width 3); prover-estimator debate (depth 2, width 3, $\rho = 2$; Brown-Cohen et al. [4]). Debate seats are balanced across items.

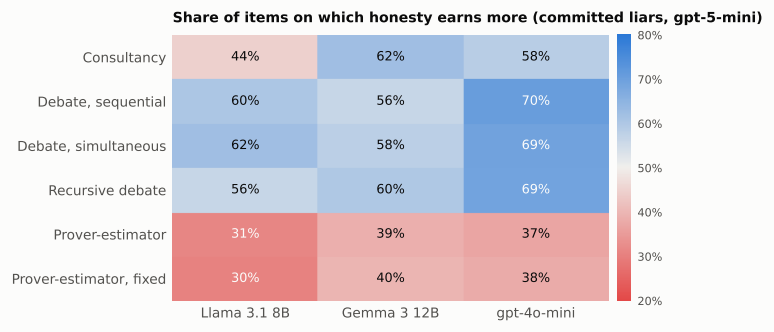


Figure 1: Share of items on which honesty earns more, per protocol and judge (committed liars, gpt-5-mini).

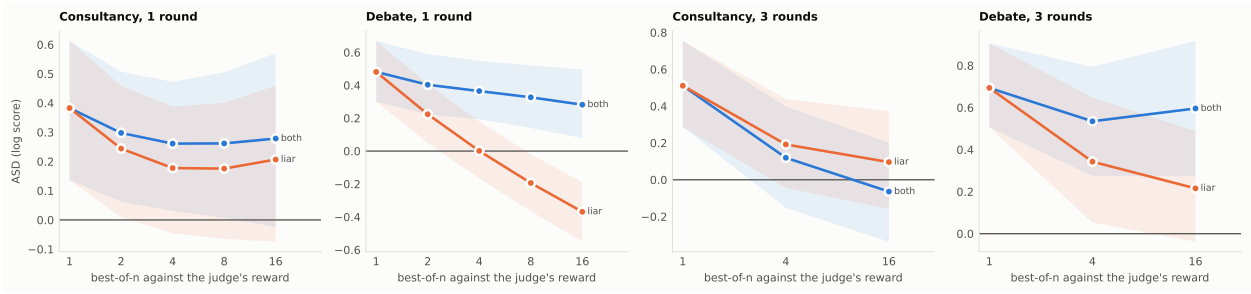


Figure 2: EIC along best-of-n, held-out judging. Blue: both sides optimized; orange: only the liar.

4 Instructed samplers

Under instructed arguments every protocol except prover-estimator pays more for honesty (Table 1). Keeping only committed liars lowers every protocol's EIC (consultancy's by almost half) but changes no sign and barely changes the ranking. Prover-estimator debate pays a prover more for defending the false answer on two-thirds of items: without a coherent estimator its subclaim pricing rewards lying. Figure 1 shows the same pattern for each judge; consultancy under Llama-3.1-8B is already negative (instructed ASD -0.26 [-0.41, -0.12]).

The right zero point on these items is not zero. A blind judge, shown no argument, pays -0.58 [-0.77, -0.40] for honesty in log score (-0.11 in probability; honesty wins on 43% of items), because the items were selected for the judges' prior being wrong. Every protocol except prover-estimator is far above that line under every sampler, including consultancy after optimization: arguments move the judge off its prior even where optimizing them no longer favours honesty.

5 Optimization pressure

Best-of-n moves the agent along Eq. (1)'s path, so the change of EIC with n shows whether the protocol stays incentive compatible as optimization proceeds (Figure 2). In one round, optimizing only the liar reverses debate's incentive: at best-of-16 the ASD is -0.37 [-0.55, -0.19] and honesty wins on 29% of items. Optimizing both sides keeps it positive ($+0.28$ [$+0.08$, $+0.50$]). With three rounds and best-of-16 on every speech, debate keeps a positive incentive with both sides optimized ($+0.60$ [$+0.27$, $+0.92$]), and with only the liar optimized it falls to a third of its starting value ($+0.22$ [-0.04, $+0.49$]); consultancy's falls to zero either way (-0.06 and $+0.10$). Extra rounds let the honest debater rebut a better lie, which consultancy's judge cannot.

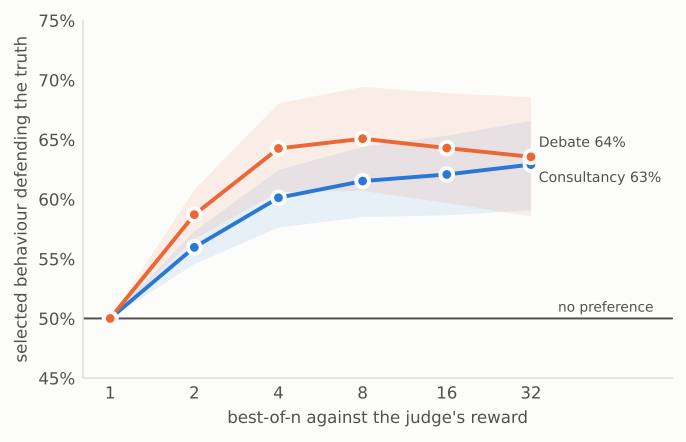


Figure 3: An agent that chooses which answer to defend, optimized by best-of-n over both sides' candidates: share of its selected behaviour that is honest (one round, held-out).

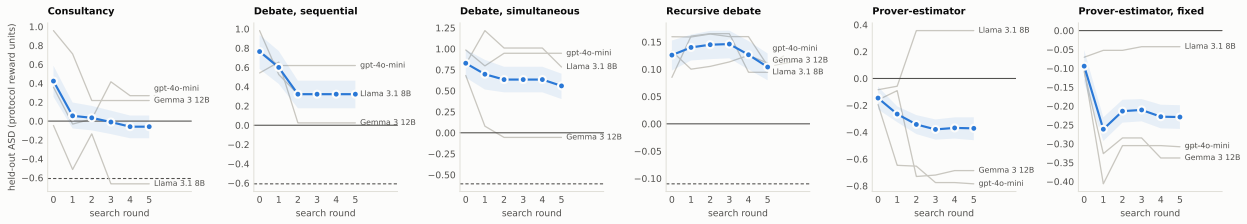


Figure 4: EIC along prompt optimization: held-out ASD of the best honest vs best deceptive strategy after each search round (gpt-5-mini; blue: judges pooled; gray: each judge).

When the agent may choose its side, best-of-n selects honest behaviour more often: from 50% to 62% [59, 65] (consultancy) and 64% [60, 69] (debate) at $n = 16$ (Figure 3). The push toward honesty is real but weak, and saturates by $n = 8$.

6 Prompt optimization

Prompt optimization is the strongest sampler here (Figure 4, Table 2). Every protocol's EIC falls as the searches proceed, because deception improves faster than honest argument. Consultancy's incentive is erased (from +0.42 to −0.06); under Llama-3.1-8B the best deceptive consultant outearns the best honest one (−0.66 [-0.91, -0.44]). Prover-estimator's deficit deepens (§11). Debate's EIC shrinks by 58% (sequential) and 33% (simultaneous) but stays significantly positive when pooled over judges; under Gemma-3-12B both debates end near zero. Recursive debate's small positive incentive barely moves. Over every strategy the searches produced (a population covariance rather than a frontier), the honest-minus-deceptive reward is +0.41 [+0.11, +0.72] for sequential debate, +0.50 [+0.16, +0.84] for simultaneous debate and +0.16 [-0.08, +0.40] for consultancy.

Figure 5 shows why. Both searches climb on their training items, but only the deceptive search's gains transfer. Over five rounds, the best deceptive strategy's held-out reward rises by +0.25 [+0.10, +0.39] (mean over the 15 protocol–judge cells with the fixed prover-estimator, bootstrap over cells; positive in 12), the best honest one's by −0.02 [-0.13, +0.07]. In both debates the honest strategies chosen on training reward do worse on held-out items than the unprompted agent (−0.19 and −0.20): with a judge whose verdicts on the same argument correlate at $r \approx 0.2$, an

Table 2: EIC under prompt optimization (held-out, 3 judges pooled): round 0 is the unprompted agent on each side; round 5 the best strategies after five rounds of search. 480 held-out items (80 for the original prover-estimator).

protocol	round 0	round 5	concordance, round 5
consultancy	+0.42	−0.06 [-0.18, +0.06]	0.47
debate, sequential	+0.77	+0.32 [+0.18, +0.46]	0.65
debate, simultaneous	+0.83	+0.56 [+0.40, +0.70]	0.67
recursive debate	+0.13	+0.10 [+0.08, +0.13]	0.60
prover-estimator, original	−0.15	−0.37 [-0.46, -0.29]	0.16
prover-estimator, fixed	−0.09	−0.23 [-0.26, -0.20]	0.27
blind judge (log score)	−0.61 [-0.88, -0.33]		0.45

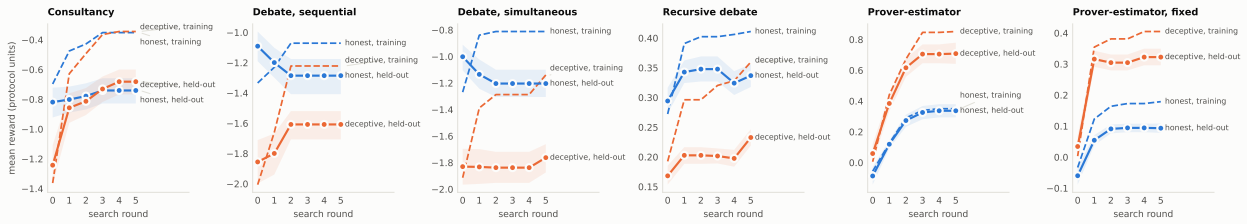


Figure 5: Goodhart curves of the prompt search: reward of each search's best strategy so far on the 36 training items it was selected on (dashed) and on 480 held-out items (solid, 95% item-bootstrap band); gpt-5-mini, judges pooled.

honest search mostly fits judge noise, while deceptive strategies find something that generalizes. Prover-estimator debate is the exception, where both searches gain (fixed payoff: honest +0.15, deceptive +0.29).

7 Self-play equilibria

The prompt searches optimize each side against the unprompted other side, so they cannot separate "debate rewards honesty at equilibrium" from "the honest side was under-optimized". We therefore ran Policy-Space Response Oracles [8] with prompt search as the best-response oracle. Populations of honest and lying strategies for gpt-5-mini start from the unprompted agent; every honest × liar pair plays on 100 training items (seats balanced); the bimatrix game is solved for a Nash equilibrium; each side then searches (3 rounds × 4 candidates by Claude Opus 5.5, told the opponents' strategies and weights) for a best response to the other's equilibrium mix, for four iterations. The judge's verdict is the mean of three samples, so that selection is not dominated by judge noise. Every pair is re-played on 390 held-out items, and EIC at iteration k is the held-out $\mathbb{E}_{\sigma_k}[u_{\text{honest}} - u_{\text{liar}}]$.

Simultaneous debate stays incentive compatible at every equilibrium and under every judge (Figure 6): pooled ASD +0.54 at iteration 0 and +0.48 [+0.38, +0.58] at iteration 4 (gpt-4o-mini +0.58, Gemma +0.55, Llama +0.31). Sequential debate does not: pooled ASD falls from +0.46 to +0.05 [-0.04, +0.13], and under gpt-4o-mini the equilibrium pays lying more (−0.19 [-0.35, -0.02]). The strategies are legible. Liars converge on fabricated, unfalsifiable specifics (named studies, benchmark figures) and, faced with arithmetic they cannot dispute, accept it and attack its applicability; the honest strategies that answer them solve the question first, present the simplest checkable object, and warn the judge that "a correct argument never needs" uncheckable citations.

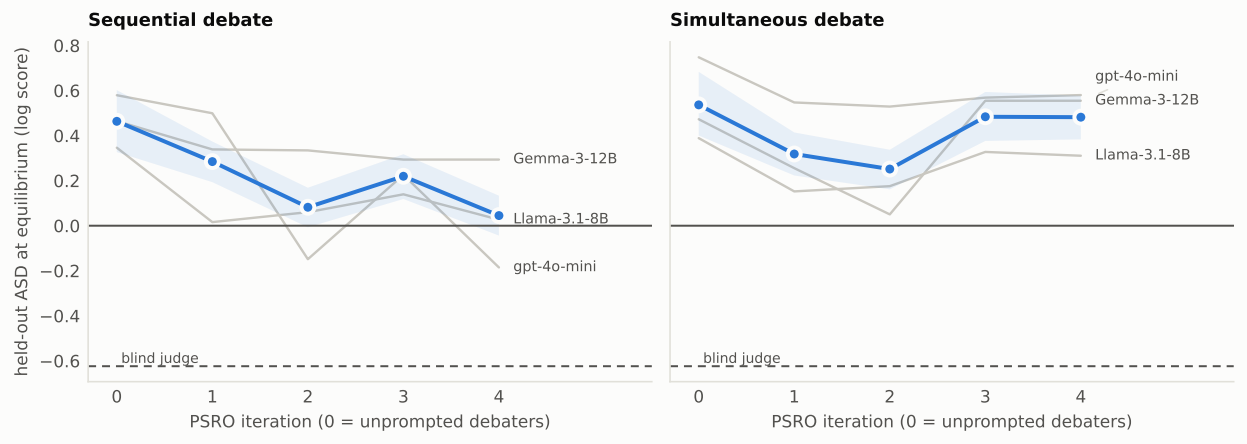


Figure 6: Held-out ASD at each PSRO iteration's equilibrium (390 items). Blue: three judges pooled, 95% item-bootstrap band; gray: each judge; dashed: the blind judge.

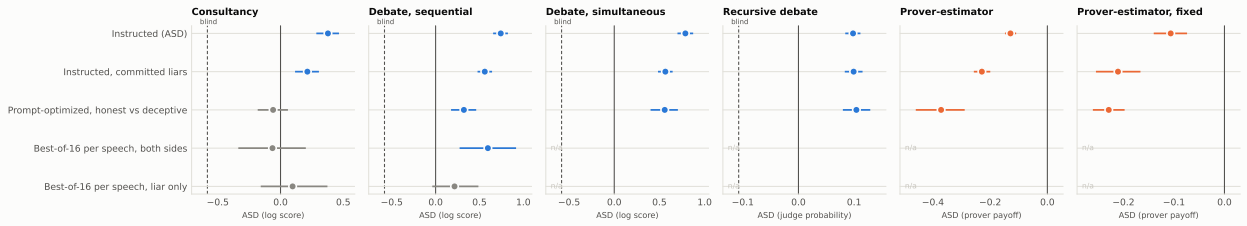


Figure 7: EIC (ASD) of each protocol under each sampler, gpt-5-mini, judges pooled. Blue: interval above zero; orange: below; gray: overlaps zero. Dashed: the blind judge, shown no argument.

As in the one-sided searches, liar best responses transfer to held-out items (7 of 14 significantly, mean held-out gain +0.09) more than honest ones (5 of 18, mean −0.01). Four iterations have not converged: sequential debate under gpt-4o-mini cycles, and some best responses still improved at iteration 4.

8 Do the samplers agree?

Figure 7 collects the samplers. EIC is measured to predict what an optimizer of the protocol's reward would do, and prompt optimization is our closest such optimizer; the question is whether the cheap instructed measurement could stand in for it. We compute concordance both ways for each of the 15 (protocol, judge) cells and compare the orderings. The instructed sampler and its committed-liar variant rank the 15 (protocol, judge) cells almost identically (Spearman 0.99), but either ranks them only partly like prompt optimization does (0.69 across cells with the fixed prover-estimator, 0.39 with the original; within a judge, over only five protocols, 0.20–0.60; Figure 8). Instructed ASD gets the extremes right on average—prover-estimator is not incentive compatible, debate is—but misses which protocols survive optimization under a given judge. Under Gemma-3-12B it rates consultancy the most incentive-compatible protocol (+0.91 [+0.76, +1.06], against +0.64 and +0.68 for the debates); after optimization all three are near zero (+0.22, +0.02, −0.06). Under gpt-4o-mini, simultaneous debate's incentive instead grows with optimization (+0.83 to +0.95).

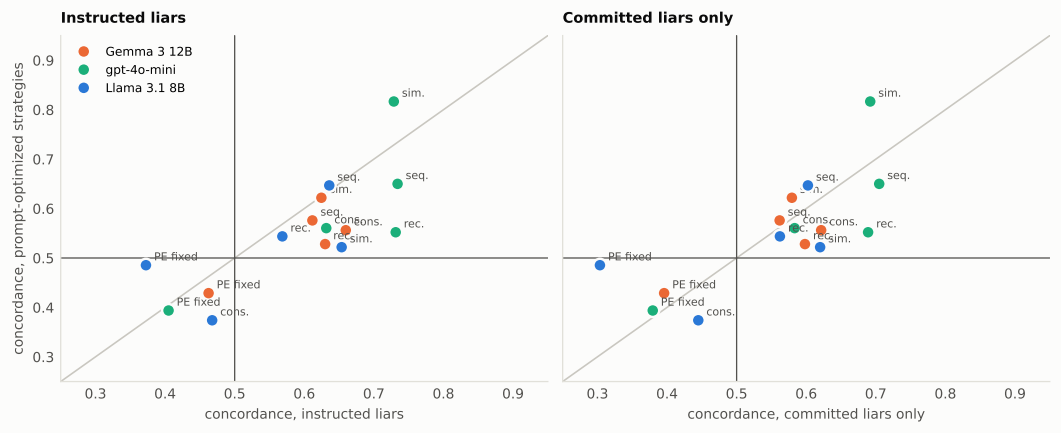


Figure 8: Concordance under the instructed samplers vs prompt optimization, one point per (protocol, judge).

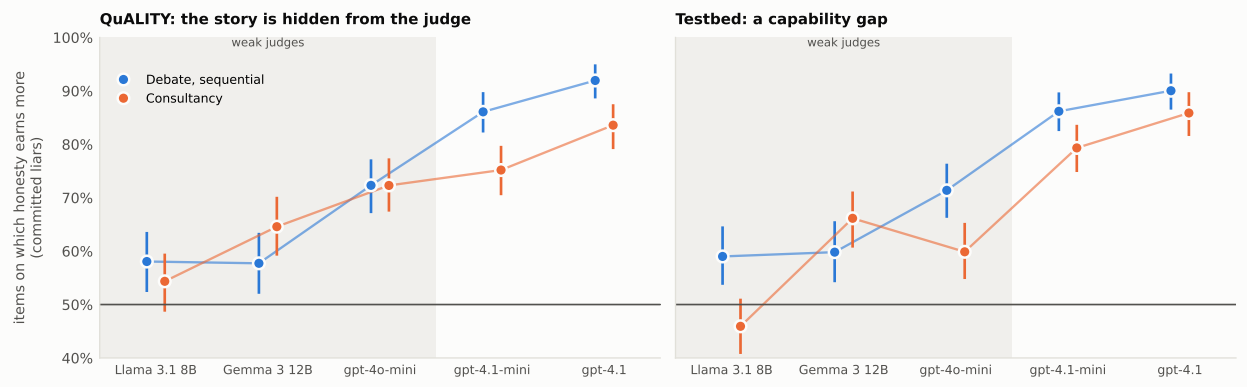


Figure 9: Concordance of the same debates and consultancies judged by five judges (committed liars).

9 EIC and judge capability

Debate transcripts do not depend on the judge, so we re-judged every stored debate with two stronger judges (gpt-4.1-mini, gpt-4.1), replaying the debaters' speeches exactly, on our testbed and on QuALITY [11] (the story hidden from the judge). EIC rises steeply with judge capability (Figure 9): in debate, honesty wins on 58–61% of items with Llama-3.1-8B and 90–92% with gpt-4.1, on both datasets. On the testbed the stronger judges already answer 87–90% of items correctly alone, and committed liars *lower* their accuracy by 12–22 points, yet the protocols' incentive under them is the highest we measured. Accuracy and incentive compatibility dissociate, as the argument for measuring incentives predicts.

10 Liars that refuse

In a tournament of eight debater models (Figure 10), the per-model EIC with prompted liars ranges from +0.74 (gpt-5) to +3.98 (Claude Opus 5.5). The spread is almost entirely refusal: Opus 5.5 stays committed in 28% of its lies (and refuses outright in 446 of 600 lying consultancies), gpt-5 in 98%. With committed liars only, every model's EIC lies between 0.0 and +0.6. Instructed samplers built from safety-trained models overstate incentive compatibility severely (Opus 5.5: +3.98 with

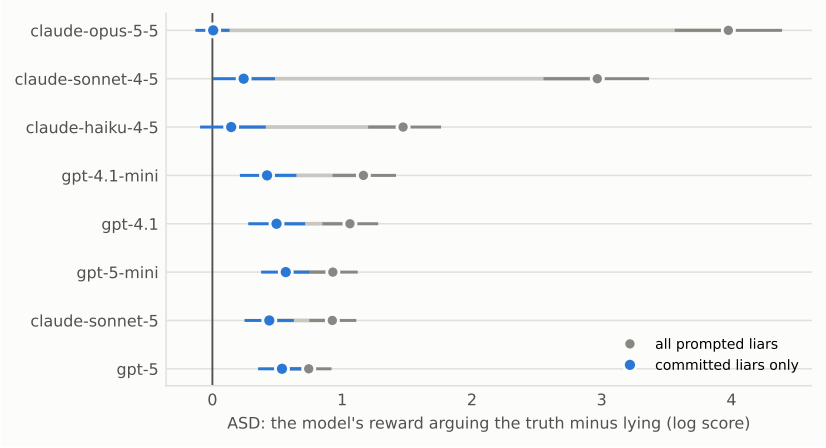


Figure 10: Per debater model: reward arguing the truth minus lying, against every opponent in an 8-model debate tournament (21,570 verdicts), all prompted liars vs committed liars only.

all prompted liars, +0.01 with committed ones); the effective-lie sampler or an optimizing sampler is needed.

11 Why prover-estimator debate fails here

Prover-estimator debate [4] pays the prover a small start term for the estimator's probability of its answer, a term per decomposition round, and a leaf term: direction $\times$ (judge's probability $-$ estimator's probability) for the final subclaim, with the direction chosen by the prover. Splitting the reward of 16,718 instructed episodes into these terms shows that the leaf term carries the deficit: the strong estimator prices true leaves at 0.87 and false ones at 0.21, while the weak judges stay near the middle (0.56 and 0.40), so an honest prover defending a true leaf loses on average (-0.20). Llama-3.1-8B is anti-informative at the leaf (0.28 for true leaves, 0.43 for false).

Our first implementation also paid each decomposition round for any gap between the estimator's probability of the claim and of the conjunction of its subclaims. Since our subclaims need only imply the claim, $P(\text{all}) < P(\text{claim})$ is coherent, and calling the claim "too high" earned money whatever the truth. The deceptive prompt search found this under every judge ("a coherent estimator always gives $P(\text{all}) \leq P(\text{claim})$, so say too_high every round"), adding logically impossible decoy subclaims. We fixed the payoff to pay only for the incoherence a sufficient decomposition can show, $P(\text{claim}) < P(\text{all})$, and re-ran both samplers. The fix removes the exploit but not the deficit: instructed ASD -0.11 [-0.14, -0.07], prompt-optimized -0.23 [-0.26, -0.20]. The deceptive search now finds a genuine lie: steer the recursion to the leaf the estimator prices lowest that still reads as true out of context, phrase it as a soft generalization ("tends to", "is more associated with"), and defend it in an authoritative voice with invented supporting observations. The leaf term rewards honesty only if the leaf judge is more accurate than the estimator; with a weak judge and a strong estimator it rewards finding where the judge is most wrong.

12 Hazards in measuring incentives against LLM judges

Judge noise. Two independent samples of the same judge on the same argument correlate $r =$ 0.08–0.33 within an item. Any sampler that selects on one judge sample and scores on the same

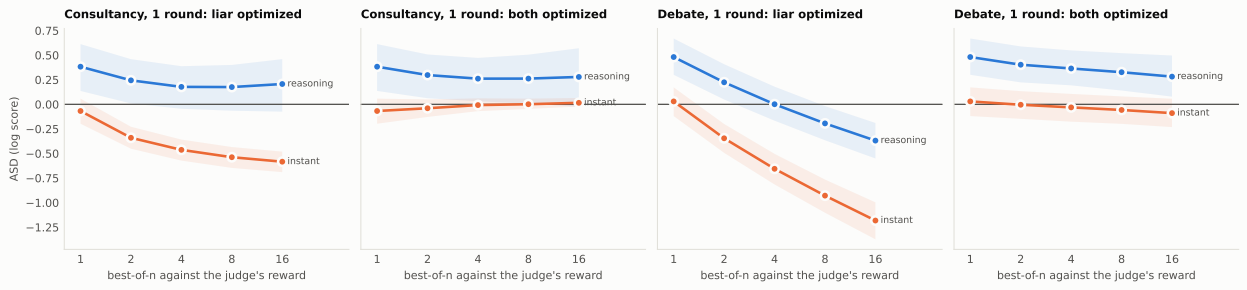


Figure 11: ASD along best-of-n (one round, gpt-5-mini, judges pooled): instant judges (orange) vs the same judges reasoning first, scored on independent samples (blue).

one inflates the effect of optimization (in-sample best-of-16 consultancy liars appear to push the judge to 0.36 accuracy; held out, 0.55). Covariance estimates are attenuated by the same noise.

Reward scale. Log-score rewards grow with the judge's confidence, so ASD in log-score units is not comparable across judges; concordance and r are.

Last-word bias. Weak judges side with the debater who speaks second: in sequential debate the honest side wins 0.75 of the time speaking second and 0.42 speaking first. Seat balancing is necessary for any EIC estimate of sequential protocols.

Instant judges. Earlier best-of-n studies scored arguments with judges that answer at once. We re-judged the stored transcripts and best-of-16 pools with each weak judge answering without reasoning at temperature 0 (two calls on one transcript correlate at $r = 0.78$–0.95). Instructed EIC falls (three-round consultancy $+0.38 \to +0.00$, debates $\approx +0.77 \to +0.50$): removing the reasoning removes most of what the judge learns from the argument, not only noise. And the near-deterministic judge is far easier to exploit (Figure 11): best-of-16 liars drive one-round debate's ASD to -1.18 [-1.37, -1.00] (reasoning judge, held out: -0.37) and consultancy's to -0.58, and with both sides optimized debate gives honesty no edge at any n. The reasoning judge's noise partly protects it: an argument selected on one noisy verdict does not reliably fool the next.

Item selection. Items were kept because the weak judges alone get them wrong, so their blind baselines are negative by construction. This does not bias the within-item contrast (selection was never on a protocol's outcome, and protocol and baseline both use fresh judge samples), but it makes raw ASDs conditional on items that mislead a weak judge, and it exaggerates the rise of EIC with judge capability: the stronger judges were not selected against.

13 Discussion

The question for a training method is whether optimizing against it rewards honesty. On conceptual reasoning with weak judges, the answer depends on the protocol and on the optimizer. Instructed liars make consultancy, debate and recursive debate all look incentive compatible. A prompt optimizer that searches for good lies breaks consultancy and leaves debate's incentive positive but diminished; when both debaters best-respond to each other, simultaneous debate keeps it (+0.48) while sequential debate's is optimized away (+0.05); prover-estimator debate with a weak leaf judge

rewards lying under every sampler, with or without its payoff exploit. The incentive grows with the judge's capability, far more than with the protocol.

Limitations. v is by construction and binary; graded ground truth (e.g. expert-rated essays) would allow richer behaviour spaces. The optimizers are inference-time; whether EIC under these samplers predicts the effect of RL training is the open question this study prepares for but does not answer. The self-play runs have not converged after four iterations. The prompt-optimized liars were not audited for commitment. Judges are LLMs, not people.

Reproducibility. All experiments use the open-source `so-arena` library. The scripts are in its `experiments/` directory, the data and tables in `docs/results/`. Total API spend was about $9,160.

References

[1] Lakshya A. Agrawal et al. GEPA: Reflective prompt evolution can outperform reinforcement learning. *arXiv preprint arXiv:2507.19457*, 2025.

[2] Samuel Arnesen, David Rein, and Julian Michael. Training language models to win debates with self-play improves judge accuracy. *arXiv preprint arXiv:2409.16636*, 2024.

[3] Ahmad Beirami, Alekh Agarwal, Jonathan Berant, Alexander D'Amour, Jacob Eisenstein, Chirag Nagpal, and Ananda Theertha Suresh. Theoretical guarantees on the best-of-n alignment policy. *arXiv preprint arXiv:2401.01879*, 2024.

[4] Jonah Brown-Cohen, Geoffrey Irving, and Georgios Piliouras. Avoiding obfuscation with prover-estimator debate. *arXiv preprint arXiv:2506.13609*, 2025.

[5] Geoffrey Irving, Paul Christiano, and Dario Amodei. AI safety via debate. *arXiv preprint arXiv:1805.00899*, 2018.

[6] Zachary Kenton, Noah Y. Siegel, János Kramár, Jonah Brown-Cohen, Samuel Albanie, Jannis Bulian, Rishabh Agarwal, David Lindner, Yunhao Tang, Noah D. Goodman, and Rohin Shah. On scalable oversight with weak LLMs judging strong LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[7] Akbir Khan, John Hughes, Dan Valentine, Laura Ruis, Kshitij Sachan, Ansh Radhakrishnan, Edward Grefenstette, Samuel R. Bowman, Tim Rocktäschel, and Ethan Perez. Debating with more persuasive LLMs leads to more truthful answers. In *International Conference on Machine Learning (ICML)*, 2024.

[8] Marc Lanctot, Vinicius Zambaldi, Audrunas Gruslys, Angeliki Lazaridou, Karl Tuyls, Julien Pérolat, David Silver, and Thore Graepel. A unified game-theoretic approach to multiagent reinforcement learning. In *NeurIPS*, 2017.

[9] Julian Michael, Salsabila Mahdi, David Rein, Jackson Petty, Julien Dirani, Vishakh Padmakumar, and Samuel R. Bowman. Debate helps supervise unreliable experts. *arXiv preprint arXiv:2311.08702*, 2023.

[10] Abhimanyu Pallavi Sudhir, Jackson Kaunismaa, and Arjun Panickssery. A benchmark for scalable oversight mechanisms. *arXiv preprint arXiv:2504.03731*, 2025.

[11] Richard Yuanzhe Pang, Alicia Parrish, Nitish Joshi, Nikita Nangia, Jason Phang, Angelica Chen, Vishakh Padmakumar, Johnny Ma, Jana Thompson, He He, and Samuel R. Bowman. QuALITY: Question answering with long input texts, yes! In *NAACL*, 2022.

[12] Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. *arXiv preprint arXiv:2309.03409*, 2023.

A Judge accuracy (secondary)

For completeness: with committed liars, consultancy, debate and recursive debate raise the weak judges' accuracy by 11–14 points over the judge alone (about a fifth of the 54-point gap), prover-estimator by 5. Under information asymmetry (QuALITY) the weak judges gain 0–16 points from debate and the gpt-4.1 judges 25–27. Full tables are in `docs/results/decisive.md` and `decisive_extended.md`.